

Title

Mobile App-Delivered Respiratory Rehabilitation in Chronic Lung Disease: Efficacy and Implementation in Primary Care Settings

Abstract

Ninety participants with chronic respiratory conditions were randomised to receive either a 12-week smartphone application-guided rehabilitation curriculum or conventional outpatient pulmonary management. The mobile app cohort completed supervised exercise protocols, breathing training, and educational modules accessible via their phones. Exercise testing conducted at baseline and post-intervention revealed the app-based group experienced greater improvements in capacity and objective breathing assessments compared to usual care. Functional disability scores measuring respiratory symptom burden improved significantly in the intervention arm. A substantial proportion of app users (approximately 80%) rated the interface as intuitive, and over 60% reported perceived benefit for breathing comfort. However, adherence was variable, with fewer than half completing all prescribed sessions. Health expenditures between groups remained comparable. Feasibility implementation in four community primary care clinics demonstrated similar benefit profiles. No treatment-related injuries or disease flare-ups were documented. These findings support the effectiveness of app-based pulmonary rehabilitation for improving clinical outcomes, particularly in motivated and adherent participants. Future efforts should address strategies to sustain long-term engagement with digital rehabilitation platforms.